# Exam

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Autonomous Driving System

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# Exam Format

- Project + Oral part.
- The project is individual.
- No due date, book an appointment when ready.
- The oral part will be done on the same day of the project discussion.

# Project: Bare Minimum

- Extend the proposed BA notebook in order to
  - Change the dataset with one with distorted images and handle it
    - TUM dataset
    - MUN-FRL dataset
    - EURO-MAV dataset
    - Search online for Visual-Odometry Dataset
  - Reconstruct the environment with at least 100 frames
    - Triangulate new points from previous images when the number of available 3D points for PnP fell under a certain threshold
- The exam will also have an oral part on the theory of the course

# Project: Medium

- All bare minimum points plus one or more of these:
  - Change feature extractor and/or optical flow algorithm (i.e. use neural network)
  - Change the BA formulation to estimate also the intrinsics parameters of the camera (both $\mathbf{K}$ and $\mathbf{d}$)
  - Use two cameras (stereo setup), track points in time and in common between the two cameras. Use all the 2d points to reconstruct both cameras motion in a single factor-graph.
- The exam will also have a small oral part on the theory of the course

# Project: Advanced

- All bare minimum points plus at least one medium point and one or more of these:
  - Use loop closure to re-identify lost points already seen and add reprojection factor between non-consecutive poses with common points.
  - Recover the real-world scale factor using additional GTSAM factor leveraging other sensors such as IMU, wheel odometry, gps, etc...
  - Convert the BA to a Visual Odometry. Solve the BA only on the last N frames ensuring continuity in the estimated poses.
- The exam will also have a small oral part in which the student must example a little bit of theory behind the proposed advance points implemented.